Exercise – Understanding simultaneity bias using simulation

1. Aim of the exercise

The aim of this exercise is to illustrate how simultaneity generates endogeneity through simulation.

2. Theory

Consider the simultaneous equations model

$$\begin{aligned} y_{1i} &= \beta_1 y_{2i} + \gamma_1 z_{1i} + u_{1i}, \\ y_{2i} &= \beta_2 y_{1i} + \gamma_2 z_{2i} + u_{2i}. \end{aligned}$$

For simplicity, the intercepts are omitted.

Assume that

$$\begin{aligned} \mathbb{E}\left[u_{1i} \mid z_{1i}, z_{2i}\right] &= 0, \\ \mathbb{E}\left[u_{2i} \mid z_{1i}, z_{2i}\right] &= 0. \end{aligned}$$

By the law of iterated expectations, these assumptions imply

$$\begin{aligned} \mathbb{E}\left[u_{1i}\right] &= 0, \\ \mathbb{E}\left[u_{2i}\right] &= 0. \end{aligned}$$

They also imply

$$\mathbb{E}\left[z_{ji} u_{ki}\right] = 0,\ j, k \in \{1, 2\},$$

so that both exogenous variables, z_{1i} and z_{2i}, are uncorrelated with both disturbance terms, u_{1i} and u_{2i}.

Suppose that the parameter of interest is β_1 in the first equation,

$$y_{1i} = \beta_1 y_{2i} + \gamma_1 z_{1i} + u_{1i}.$$

To determine whether β_1 can be estimated consistently by OLS, we must examine whether the regressor y_{2i} is uncorrelated with the disturbance term u_{1i}.

To determine whether y_{2i} is correlated with u_{1i}, we solve the system for y_{2i} in terms of the exogenous variables and the disturbance terms.

Substituting the first equation into the second equation gives

$$\begin{aligned} y_{2i} &= \beta_2 y_{1i} + \gamma_2 z_{2i} + u_{2i} \\ &= \beta_2 \left(\beta_1 y_{2i} + \gamma_1 z_{1i} + u_{1i}\right) + \gamma_2 z_{2i} + u_{2i} \\ &= \beta_1 \beta_2 y_{2i} + \gamma_1 \beta_2 z_{1i} + \beta_2 u_{1i} + \gamma_2 z_{2i} + u_{2i}. \end{aligned}$$

Collecting the terms involving y_{2i} yields

$$\left(1 - \beta_1 \beta_2\right) y_{2i} = \gamma_1 \beta_2 z_{1i} + \gamma_2 z_{2i} + \beta_2 u_{1i} + u_{2i}.$$

Assuming that

$$1 - \beta_1 \beta_2 \neq 0,$$

we obtain the reduced-form expression

$$y_{2i} = z_{1i}\frac{\gamma_1\beta_2}{1-\beta_1\beta_2} + z_{2i}\frac{\gamma_2}{1-\beta_1\beta_2} + u_{1i}\frac{\beta_2}{1-\beta_1\beta_2} + u_{2i}\frac{1}{1-\beta_1\beta_2}.$$

Solving the system for y_{1i} in the same way yields

$$\begin{aligned}
y_{1i} &= \beta_1 y_{2i} + \gamma_1 z_{1i} + u_{1i} \\
&= \beta_1\left(\frac{\beta_2\gamma_1 z_{1i} + \gamma_2 z_{2i} + \beta_2 u_{1i} + u_{2i}}{1-\beta_1\beta_2}\right) + \gamma_1 z_{1i} + u_{1i} \\
&= \frac{\beta_1\beta_2\gamma_1 z_{1i} + \beta_1\gamma_2 z_{2i} + \beta_1\beta_2 u_{1i} + \beta_1 u_{2i}}{1-\beta_1\beta_2} + \gamma_1 z_{1i} + u_{1i} \\
&= \frac{\beta_1\beta_2\gamma_1 z_{1i} + \beta_1\gamma_2 z_{2i} + \beta_1\beta_2 u_{1i} + \beta_1 u_{2i} + (1-\beta_1\beta_2)\gamma_1 z_{1i} + (1-\beta_1\beta_2)u_{1i}}{1-\beta_1\beta_2} \\
&= \frac{\beta_1\beta_2\gamma_1 z_{1i} + \beta_1\gamma_2 z_{2i} + \beta_1\beta_2 u_{1i} + \beta_1 u_{2i} + \gamma_1 z_{1i} - \beta_1\beta_2\gamma_1 z_{1i} + u_{1i} - \beta_1\beta_2 u_{1i}}{1-\beta_1\beta_2} \\
&= \frac{\gamma_1 z_{1i} + \beta_1\gamma_2 z_{2i} + u_{1i} + \beta_1 u_{2i}}{1-\beta_1\beta_2}.
\end{aligned}$$

Therefore, the reduced-form representation of the simultaneous equations model is

$$y_{1i} = \frac{\gamma_1 z_{1i} + \beta_1\gamma_2 z_{2i} + u_{1i} + \beta_1 u_{2i}}{1-\beta_1\beta_2},$$

$$y_{2i} = \frac{\beta_2\gamma_1 z_{1i} + \gamma_2 z_{2i} + \beta_2 u_{1i} + u_{2i}}{1-\beta_1\beta_2}.$$

These reduced-form equations will be used in the simulation exercise to generate observations on y_{1i} and y_{2i}.

The parameter of interest is β_1 in the first equation,

$$y_{1i} = \beta_1 y_{2i} + \gamma_1 z_{1i} + u_{1i}.$$

To estimate β_1 consistently by OLS, the regressor y_{2i} must be uncorrelated with the disturbance term u_{1i}, that is,

$$\mathbb{E}\left[y_{2i}u_{1i}\right] = 0.$$

Does this condition hold?

The reduced-form expression for y_{2i} contains u_{1i} whenever $\beta_2 \neq 0$. This suggests that y_{2i} may be correlated with the disturbance term in the first equation. To verify this formally, we compute $\mathbb{E}[y_{2i}u_{1i}]$.

Consider again the reduced-form expression

$$y_{2i} = z_{1i}\frac{\gamma_1\beta_2}{1-\beta_1\beta_2} + z_{2i}\frac{\gamma_2}{1-\beta_1\beta_2} + u_{1i}\frac{\beta_2}{1-\beta_1\beta_2} + u_{2i}\frac{1}{1-\beta_1\beta_2}.$$

Multiplying both sides by u_{1i} and taking expectations yields

$$\begin{aligned}
\mathbb{E}\left[y_{2i}u_{1i}\right] = \mathbb{E}\left[z_{1i}u_{1i}\right]\frac{\gamma_1\beta_2}{1-\beta_1\beta_2} + \mathbb{E}\left[z_{2i}u_{1i}\right]\frac{\gamma_2}{1-\beta_1\beta_2} \\
+ \mathbb{E}\left[u_{1i}^2\right]\frac{\beta_2}{1-\beta_1\beta_2} + \mathbb{E}\left[u_{2i}u_{1i}\right]\frac{1}{1-\beta_1\beta_2}.
\end{aligned}$$

Since

$$\mathbb{E}\left[z_{1i}u_{1i}\right] = \mathbb{E}\left[z_{2i}u_{1i}\right] = 0,$$

it follows that

$$\mathbb{E}\left[y_{2i}u_{1i}\right] = \mathbb{E}\left[u_{1i}^2\right]\frac{\beta_2}{1-\beta_1\beta_2} + \mathbb{E}\left[u_{2i}u_{1i}\right]\frac{1}{1-\beta_1\beta_2}.$$

This shows that $\mathbb{E}\left[y_{2i}u_{1i}\right] \neq 0$ whenever either

$$\beta_2 \neq 0,$$

or

$$\mathbb{E}\left[u_{2i}u_{1i}\right] \neq 0.$$

Consider again the simultaneous equations model

$$\begin{aligned} y_{1i} &= \beta_1 y_{2i} + \gamma_1 z_{1i} + u_{1i}, \\ y_{2i} &= \beta_2 y_{1i} + \gamma_2 z_{2i} + u_{2i}. \end{aligned}$$

We have shown that

$$\mathbb{E}\left[y_{2i}u_{1i}\right] \neq 0.$$

Therefore, in the first equation, the regressor y_{2i} is correlated with the disturbance term u_{1i}. The exogeneity assumption required for OLS is violated and, consequently, the OLS estimator of β_1 is generally biased and inconsistent. This phenomenon is known as simultaneity bias.

What is the implication of

$$\mathbb{E}\left[y_{2i}u_{1i}\right] \neq 0$$

for the OLS estimator of β_1? The OLS estimator from the first equation is

$$\hat{\beta}_1 = \left(\sum_{i=1}^n y_{2i}^2\right)^{-1} \sum_{i=1}^n y_{2i}y_{1i}.$$

Substituting the first structural equation,

$$y_{1i} = \beta_1 y_{2i} + \gamma_1 z_{1i} + u_{1i},$$

yields

$$\begin{aligned} \hat{\beta}_1 &= \left(\sum_{i=1}^n y_{2i}^2\right)^{-1} \sum_{i=1}^n y_{2i}\left(\beta_1 y_{2i} + \gamma_1 z_{1i} + u_{1i}\right) \\ &= \beta_1 + \left(\sum_{i=1}^n y_{2i}^2\right)^{-1} \sum_{i=1}^n y_{2i}\gamma_1 z_{1i} + \left(\sum_{i=1}^n y_{2i}^2\right)^{-1} \sum_{i=1}^n y_{2i}u_{1i}. \end{aligned}$$

Since

$$\mathbb{E}\left[y_{2i}u_{1i}\right] \neq 0,$$

it follows that

$$\mathbb{E}\left[u_{1i} \mid y_{2i}, z_{1i}\right] \neq 0.$$

Taking expectations conditional on y_{2i} and z_{1i}, we have

$$\mathbb{E}\left[\hat{\beta}_1 \mid y_{2i}, z_{1i}\right] = \beta_1 + \left(\sum_{i=1}^{n} y_{2i}^2\right)^{-1} \sum_{i=1}^{n} y_{2i}\gamma_1 z_{1i} + \left(\sum_{i=1}^{n} y_{2i}^2\right)^{-1} \sum_{i=1}^{n} y_{2i}\mathbb{E}\left[u_{1i} \mid y_{2i}, z_{1i}\right].$$

Since

$$\mathbb{E}\left[u_{1i} \mid y_{2i}, z_{1i}\right] \neq 0,$$

the final term is generally nonzero. Therefore,

$$\mathbb{E}\left[\hat{\beta}_1 \mid y_{2i}, z_{1i}\right] \neq \beta_1.$$

Thus, the OLS estimator of β_1 is biased. Since y_{2i} is determined simultaneously with y_{1i}, it is correlated with the disturbance term u_{1i}, violating the exogeneity assumption required for unbiased OLS estimation. This phenomenon is known as simultaneity bias.

3. Generate data from a simultaneous equations model

We start by simulating data from a simultaneous equations model to illustrate how simultaneity generates endogeneity. A sample of 1,000 observations is generated by assuming values for the structural parameters and drawing the exogenous variables and structural disturbances from their respective distributions. The endogenous variables are then obtained from the reduced-form solution of the system.

To demonstrate the violation of the exogeneity assumption, the sample correlation between y_2 and the structural disturbance u_1 is computed. Since the reduced-form expression for y_2 contains a component of u_1, the correlation is expected to be nonzero. This provides a numerical illustration of the fact that an endogenous regressor is correlated with the error term, which leads to inconsistent OLS estimation in simultaneous equations models.

```
%% 4. Simulating a simultaneous equations model

% 4.1. Set sample size
N_obs = 1000;

% 4.2. Set structural parameters
B_1 = 0.5;
B_2 = 0.6;
g_1 = 1.0;
g_2 = 1.0;

% 4.3. Generate exogenous variables
z_1 = random('Normal',0,1,[N_obs 1]);
z_2 = random('Normal',0,1,[N_obs 1]);

% 4.4. Generate structural errors
u_1 = random('Normal',0,1,[N_obs 1]);
u_2 = random('Normal',0,1,[N_obs 1]);
```

```
36
37 % 4.5. Solve the simultaneous system
38 y_1 = (g_1 * z_1 + B_1 * g_2 * z_2 + u_1 + B_1 * u_2) / ...
39     (1 - B_1 * B_2);
40 y_2 = (B_2 * g_1 * z_1 + g_2 * z_2 + B_2 * u_1 + u_2) / ...
41     (1 - B_1 * B_2);
42
43 % 4.6. Compute the correlation between y_2 and u_1
44 corr_y_2_u_1 = corr(y_2,u_1);
```

4. OLS estimation of the first structural equation

We estimate the first structural equation using OLS. We construct the regressor matrix from the endogenous variable y_2 and the exogenous variable z_1, corresponding to the specification of the first structural equation. We then estimate the coefficients of the model using OLS. The particular interest lies in the estimate of β_1, which measures the effect of y_2 on y_1. Since y_2 is correlated with the structural disturbance u_1, this coefficient estimate is expected to be biased and inconsistent. This illustrates the simultaneity bias that arises when an endogenous variable is treated as an exogenous regressor.

```
46 %% 5. OLS estimation of the first structural equation
47
48 % Create the systematic part of the regression
49 X = [y_2 z_1];
```

5. Sampling distribution of the OLS estimator

We conduct a Monte Carlo simulation to investigate the finite-sample properties of the OLS estimator. For each of 1,000 replications, we first generate new realizations of the exogenous variables and structural disturbances. We then solve the simultaneous equations system to obtain the endogenous variables. The first structural equation is then estimated by OLS, and the resulting estimate of β_1 is stored. Repeating this procedure many times produces an empirical sampling distribution for $\hat{\beta}_1$. We can use this distribution to examine the estimator's behavior and to assess the effects of simultaneity bias.

```
51 %% 6. Sampling distribution of the OLS estimator
52
53 % 6.1. Set the number of simulations
54 N_sim = 1000;
55
56 % 6.2. Preallocate a vector for the simulated estimates of B_1
57 B_hat_1_sim = NaN(N_sim,1);
58
59 % 6.3. Create the sampling distribution
60 for i = 1:N_sim
61     z_1 = random('Normal',0,1,[N_obs 1]);
62     z_2 = random('Normal',0,1,[N_obs 1]);
63     u_1 = random('Normal',0,1,[N_obs 1]);
64     u_2 = random('Normal',0,1,[N_obs 1]);
```

```
    y_1 = (g_1 * z_1 + B_1 * g_2 * z_2 + u_1 + B_1 * u_2) / ...
        (1 - B_1 * B_2);
    y_2 = (B_2 * g_1 * z_1 + g_2 * z_2 + B_2 * u_1 + u_2) / ...
        (1 - B_1 * B_2);
    X = [y_2 z_1];
    LSS = lss(y_1,X);
    B_hat_1_sim(i,1) = LSS.B_hat(1,1);
end
```

6. Bias of the OLS estimator

We evaluate the bias of the OLS estimator using the empirical sampling distribution obtained from the Monte Carlo simulation. First, we compute the mean of the sampling distribution of $\hat{\beta}_1$. We then calculate the bias as the difference between the average estimated value and the true parameter value, β_1. If OLS were unbiased, this difference would be close to zero. However, because y_2 is endogenous and correlated with the structural disturbance u_1, the estimated bias is expected to be nonzero, confirming that the OLS estimator is affected by simultaneity bias.

```
%% 7. Bias of the OLS estimator

% 7.1. Compute the mean of the sampling distribution
B_hat_1_mean = mean(B_hat_1_sim);

% 7.2. Compute the bias
bias = B_hat_1_mean - B_1;
```

7. Plot the sampling distribution of the OLS estimator

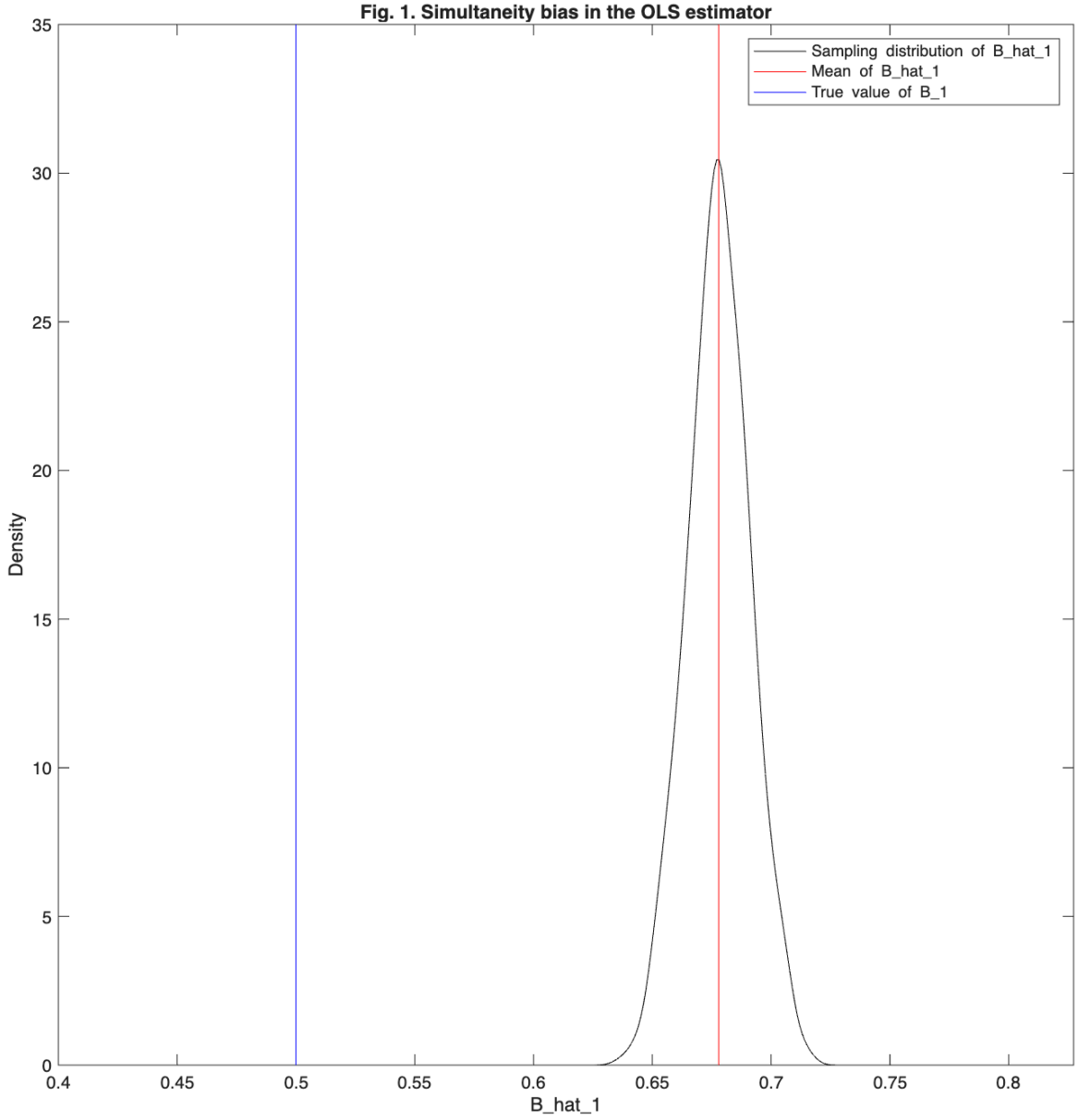
We visualize the empirical sampling distribution of $\hat{\beta}_1$ using a kernel density estimate based on the Monte Carlo replications. The resulting density provides a graphical representation of the variability of the OLS estimator across repeated samples. We plot the mean of the sampling distribution alongside the true parameter value to assess the presence of simultaneity bias. If OLS were unbiased, the center of the sampling distribution would coincide with the true parameter value. Any systematic deviation between the two indicates bias arising from the endogeneity of y_2. The figure illustrates the bias induced by the endogeneity of y_2.

```
%% 8. Plot the sampling distribution of the OLS estimator

% Create a figure
figure('Position',[100 100 1000 1000]);
% Estimate the density
[f,x] = ksdensity(B_hat_1_sim);
% Plot the density
plot(x,f,...
    'Color',[0.000 0.000 0.000],...
    'DisplayName','Sampling distribution of B\_hat\_1')
% Expand the x-axis range
lower = min([x(:);B_1]) - 0.1;
upper = max([x(:);B_1]) + 0.1;
```

```
xlim([lower upper])
hold on
% Add the mean of the sampling distribution
line([B_hat_1_mean B_hat_1_mean],ylim,...
    'Color',[1.000 0.000 0.000],...
    'DisplayName','Mean of B\_hat\_1')
% Add the true value
line([B_1 B_1],ylim,...
    'Color',[0.000 0.000 1.000],...
    'DisplayName','True value of B\_1')
title('Fig. 1. Simultaneity bias in the OLS estimator')
legend('show')
ylabel('Density')
xlabel('B\_hat\_1')
hold off
```

Fig. 1. Simultaneity bias in the OLS estimator



8. When does simultaneity bias disappear?

To illustrate the source of simultaneity bias, set the coefficient β_2 equal to zero and repeat the simulation exercise. In this case, the second endogenous variable no longer depends on the error term u_1, eliminating the correlation between y_2 and the disturbance in the first structural equation. As a result, the exogeneity condition is restored and the OLS estimator becomes unbiased. The sampling distribution of $\hat{\beta}_1$ is therefore expected to be centered much closer to the true parameter value β_1, demonstrating that simultaneity bias arises from the feedback relationship that induces correlation between the endogenous regressor and the structural error term.

9 . Final notes

This file is prepared by Tunga Kantarcı. This file and the accompanying MATLAB files are available on GitHub and can be accessed via this link.